R Programming Lab 03

Retail Sales Data Cleaning and Analysis

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1. Aim

To clean, integrate, and analyse retail transaction data using R, identify key sales and customer patterns, and store the integrated results in an SQLite database for SQL-based retrieval.

2. Tools and Packages

The experiment uses **readr** for CSV files, **jsonlite** for JSON, **readxl** and **writexl** for Excel, **dplyr** and **tidyr** for data manipulation, **DBI** and **RSQLite** for database operations, and **lubridate** for date handling.

3. Dataset and Import

The experiment uses the Online Retail dataset from the UCI Machine Learning Repository. The original data is converted into three files representing transaction records, product information, and customer information: transactions.csv, products.json, and customers.xlsx.

Source	Initial records
Transactions	541,909
Products	4,070
Customers	4,380

4. Data Cleaning

The transaction data is deduplicated, rows missing customer or product identifiers are removed, and zero or negative quantities are excluded from the sales-performance analysis. Invoice dates are converted to Date values. Product records are deduplicated by stock code, and products with missing or non-positive prices are removed. Customer records are deduplicated and entries without customer IDs or countries are excluded.

Dataset	Before	After cleaning	Removed
Transactions	541,909	392,708	149,201
Products	4,070	3,855	215
Customers	4,380	4,372	8

The cleaned transaction and product data is then integrated. An inner join excludes transactions without a valid product record, while a left join adds customer information. Revenue is calculated as Quantity × UnitPrice. The integrated dataset contains 387,877 rows and 9 columns; 4,831 transaction rows had no matching product.

5. Sales and Customer Analysis

The analysis calculates total revenue, ranks products, countries, and customers by revenue, and classifies customers into value segments according to their total purchase value.

Total revenue

Total sales revenue in the integrated dataset: **10,752,840**.

Top five products by revenue

Product	Description	Revenue
23843	PAPER CRAFT, LITTLE BIRDIE	168,470
47566	PARTY BUNTING	142,438
22423	REGENCY CAKESTAND 3 TIER	135,605
85123A	WHITE HANGING HEART T-LIGHT HOLDER	93,746
23166	MEDIUM CERAMIC TOP STORAGE JAR	81,033

Top five countries by revenue

Country	Revenue
United Kingdom	8,861,857.13
Netherlands	363,884.48
EIRE	331,660.17
Germany	263,818.97
France	226,975.60

Top five customers by revenue

Customer ID	Total revenue
18102	408,760.00
14646	357,531.10
17450	186,038.00
14911	182,690.50
16446	168,472.50

Customer value segments

Segment	Number of customers
Premium	1,085
High Value	1,738
Medium Value	1,399
Low Value	117

The highest-performing market was the United Kingdom, with revenue of 8,861,857 and 16,635 orders. The lowest-performing market in the results was Saudi Arabia, with revenue of 181.44 and one order.

6. Database Storage and SQL Queries

The integrated sales data is stored in an SQLite database named retail_analysis.sqlite, in a table called retail_sales. SQL queries are used to retrieve the five highest-revenue customers and total revenue by country.

Customer ID	Total revenue
18102	408,760.00
14646	357,531.10
17450	186,038.00
14911	182,690.50
16446	168,472.50

The SQL query results agree with the customer revenue ranking calculated during the R analysis. The country query also confirms that the United Kingdom contributes the largest share of recorded revenue.

7. Conclusion

This experiment demonstrates a complete retail data-analysis workflow in R. Data is imported from multiple file formats, cleaned according to explicit rules, combined into an integrated sales table, and analysed to identify top products, customers, and markets. Customer value segmentation provides an additional view of purchasing patterns. Finally, SQLite storage and SQL queries make the integrated data available for structured retrieval. The results show a strong concentration of revenue in the United Kingdom and highlight the importance of data quality when drawing conclusions from sales records.

Note: Zero and negative quantities are excluded from this sales-performance analysis, so returns and cancellations are not included in the reported revenue.